Lista 1 Eletromagnetismo

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1-

$$\det\left[\tilde{T}(\phi)\right] = \begin{vmatrix} \cos\phi & \operatorname{sen}\phi & 0 \\ -\operatorname{sen}\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{matrix} \cos\phi & \operatorname{sen}\phi \\ -\operatorname{sen}\phi & \cos\phi \\ 0 & 0 \end{matrix}$$

$$\cos^2 + 0 + 0 - (-\operatorname{sen}^2) + 0 + 0 = \cos^2 + \operatorname{sen}^2 = \boxed{1}$$

$$\tilde{T}(\phi)^T = \begin{vmatrix} \cos\phi & -\operatorname{sen}\phi & 0 \\ \operatorname{sen}\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{vmatrix} \equiv \tilde{T}(\phi)^{-1} = \begin{vmatrix} \cos\phi & -\operatorname{sen}\phi & 0 \\ \operatorname{sen}\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\tilde{T}(\phi)^{-1} = \begin{vmatrix} \cos\phi & \operatorname{sen}\phi & 0 \\ -\operatorname{sen}\phi & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{matrix} \cos\phi & \operatorname{sen}\phi \\ -\operatorname{sen}\phi & \cos\phi \\ 0 & 0 \end{matrix} = \begin{pmatrix} \cos & -\operatorname{sen} & 0 \\ \operatorname{sen}\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} = \tilde{T}(\phi)^{-1}$$

2-

$$\det\left[\tilde{S}(\theta)\right] = \begin{vmatrix} \operatorname{sen}\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\operatorname{sen}\theta \\ 0 & 1 & 0 \end{vmatrix} \begin{matrix} \operatorname{sen}\theta & 0 \\ \cos\theta & 0 \\ 0 & 1 \end{matrix}$$

$$\cos^2 - (-\operatorname{sen}^2) = \cos^2 + \operatorname{sen}^2 = \boxed{1}$$

$$\tilde{S}(\theta)^T = \begin{vmatrix} \operatorname{sen}\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\operatorname{sen}\theta & 0 \end{vmatrix} \equiv \tilde{S}(\theta)^{-1} = \begin{vmatrix} \operatorname{sen}\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\operatorname{sen}\theta & 0 \end{vmatrix}$$

$$S(\theta)^{-1} = \begin{vmatrix} 0 & -\operatorname{sen}\theta & \cos\theta & 0 \\ 1 & 0 & 0 & 1 \\ 0 & \cos\theta & \operatorname{sen}\theta & 0 \\ 0 & -\operatorname{sen}\theta & \cos\theta & 0 \end{vmatrix} = S(\theta)^{-1} = \begin{pmatrix} \operatorname{sen}\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\operatorname{sen} & 0 \end{pmatrix}$$

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3-

$$\det\left[T(\theta,\phi)\right] = \begin{vmatrix} \operatorname{sen}\theta\cos\phi & \operatorname{sen}\theta\operatorname{sen}\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\operatorname{sen}\phi & -\operatorname{sen}\theta \\ -\operatorname{sen}\phi & \cos\phi & 0 \end{vmatrix} \begin{matrix} \operatorname{sen}\theta\cos\phi & \operatorname{sen}\theta\operatorname{sen}\phi \\ \cos\theta\cos\phi & \cos\theta\operatorname{sen}\phi \\ -\operatorname{sen}\phi & \cos\phi \end{matrix}$$

$$0+(1\operatorname{sen}^2\theta\operatorname{sen}^2\phi)+\cos^2\theta\cos\phi-(0+(-\operatorname{sen}^2\theta\cos^2\phi)+(-\cos^2\theta\operatorname{sen}^2\phi))$$

$$+\operatorname{sen}^2\theta\operatorname{sen}^2\phi+\cos^2\theta\cos^2\phi+\operatorname{sen}^2\theta\cos^2\phi+\cos^2\theta\operatorname{sen}^2\phi =$$

$$(\cos^2\theta+\operatorname{sen}^2\theta)(\operatorname{sen}^2\phi+\cos^2\phi) = 1$$

$$\tilde{T}(\theta,\phi)^T = \begin{vmatrix} \operatorname{sen}\theta\cos\phi & \cos\theta\cos\phi & -\operatorname{sen}\phi \\ \operatorname{sen}\theta\operatorname{sen}\phi & \cos\theta\operatorname{sen}\phi & \cos\phi \\ \cos\phi & -\operatorname{sen}\theta & 0 \end{vmatrix} \equiv T(\theta,\phi)^{-1} = \begin{vmatrix} \operatorname{sen}\theta\cos\phi & \cos\theta\cos\phi & -\operatorname{sen}\phi \\ \operatorname{sen}\theta\operatorname{sen}\phi & \cos\theta\operatorname{sen}\phi & \cos\theta \\ \cos\phi & -\operatorname{sen}\theta & 0 \end{vmatrix}$$

$$T(\theta,\phi)^{-1} = \begin{vmatrix} \operatorname{sen}\theta\cos\phi & \operatorname{sen}\theta\operatorname{sen}\phi & \cos\theta & \operatorname{sen}\theta\cos\phi & \operatorname{sen}\theta\operatorname{sen}\phi \\ \cos\theta\cos\phi & \cos\theta\operatorname{sen}\phi & -\operatorname{sen}\theta & \cos\theta\cos\phi & \cos\theta\operatorname{sen}\phi \\ -\operatorname{sen}\phi & \cos\phi & 0 & -\operatorname{sen}\phi & \cos\phi \\ \operatorname{sen}\theta\cos\phi & \operatorname{sen}\theta\operatorname{sen}\phi & \cos\theta & \operatorname{sen}\theta\cos\phi & \operatorname{sen}\theta\operatorname{sen}\phi \\ \cos\theta\cos\phi & \cos\theta\operatorname{sen}\phi & -\operatorname{sen}\theta & \cos\theta\cos\phi & \cos\theta\operatorname{sen}\phi \end{vmatrix}$$

$$T(\theta,\phi)^{-1} = \begin{pmatrix} d_1 & d_4 & d_7 \\ d_2 & d_5 & d_8 \\ d_3 & d_6 & d_9 \end{pmatrix} = \begin{pmatrix} \operatorname{sen}\theta\cos\phi & \cos\theta\cos\phi & -\operatorname{sen}\phi \\ \operatorname{sen}\theta\operatorname{sen}\phi & \cos\theta\operatorname{sen}\phi & \cos\theta \\ \cos\phi & -\operatorname{sen}\theta & 0 \end{pmatrix}$$

4- $\vec{A} = x^2\cos\theta\,\hat{a}_R + 2yz\operatorname{sen}\phi\,\hat{a}_\theta$ $\qquad \vec{x} = -2\hat{a}_x + 3\hat{a}_y + 4\hat{a}_z$

$$\vec{A} = x^2\cos\theta(\operatorname{sen}\theta\cos\phi\,\hat{a}_x + \operatorname{sen}\theta\operatorname{sen}\phi\,\hat{a}_y + \cos\theta\,\hat{a}_z) +$$
$$2yz\operatorname{sen}\phi(\cos\theta\cos\phi\,\hat{a}_x + \cos\theta\operatorname{sen}\phi\,\hat{a}_y - \operatorname{sen}\theta\,\hat{a}_z)$$

$$\vec{A} = (x^2\cos\theta\operatorname{sen}\theta\cos\phi + 2yz\cos\theta\operatorname{sen}\phi\cos\phi)\,\hat{u}_x +$$
$$(x^2\cos\theta\operatorname{sen}\theta\operatorname{sen}\phi + 2yz\cos\theta\operatorname{sen}^2\phi)\,\hat{a}_y +$$
$$(x^2\cos^2\theta - 2yz\operatorname{sen}\phi\operatorname{sen}\theta)\,\hat{a}_z$$

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$$\vec{A} = \cos\theta(x^2 \operatorname{sen}\theta \cos\phi + 2yz \operatorname{sen}\theta \cos\phi)\hat{a}_x + \cos\theta \operatorname{sen}\phi (x^2 \operatorname{sen}\theta + 2yz \operatorname{sen}\phi)\hat{a}_y + (x^2 \cos^2\theta - 2yz \operatorname{sen}\phi)\hat{a}_z$$

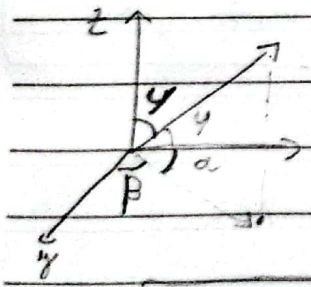
Usando as relações: $\cos\theta = \frac{z}{R}$, $\operatorname{sen}\theta = \frac{r}{R}$, $\cos\phi = \frac{x}{r}$, $\operatorname{sen}\phi = \frac{y}{r}$, $r^2 = \sqrt{x^2+y^2}$, $R = \sqrt{x^2+y^2+z^2}$, temos,

$$\vec{A} = \frac{z}{R}\left(\frac{x^3}{R} + \frac{2xy^2z}{r^2}\right)\hat{a}_x + \frac{yz}{R}\left(\frac{x^2}{R} + \frac{2y^2z}{r^2}\right)\hat{a}_y + \left(\frac{x^2z^2}{R^2} - \frac{2y^2z}{R^2}\right)\hat{a}_z$$

$$\vec{A} = \frac{4}{\sqrt{29}}\left(\frac{-8}{\sqrt{29}} + \frac{-144}{77}\right)\hat{a}_x + \frac{72}{\sqrt{29}}\left(\frac{4}{\sqrt{29}} + \frac{72}{77}\right)\hat{a}_y + \left(\frac{64}{29} - \frac{72}{77}\right)\hat{a}_z$$

$$\boxed{\vec{A} = -9{,}33\,\hat{a}_x + 74\,\hat{a}_y - 77{,}76\,a_z}$$

5- $\vec{A}$, pode ser decomposto como $\vec{A} = A\hat{a}_x + A\hat{a}_y + A\hat{a}_z$



, temos a seguinte relação trigonométrica:

$$\begin{cases} \cos^2(\alpha) = \left(\frac{A_x}{|\vec{A}|}\right)^2 \\ \cos^2(\beta) = \left(\frac{A_y}{|\vec{A}|}\right)^2 \\ \cos^2(\gamma) = \left(\frac{A_z}{|\vec{A}|}\right)^2 \end{cases}$$

$|\vec{A}| = \sqrt{A\hat{a}_x^2 + A\hat{a}_y^2 + A\hat{a}_z^2}$, substituindo as relações temos:

$$\frac{A_x^2}{(A\hat{a}_x + A\hat{a}_y + A\hat{a}_z)^2} + \frac{A_y^2}{(A\hat{a}_x + A\hat{a}_y + A\hat{a}_z)^2} + \frac{A_z^2}{(A\hat{a}_x + A\hat{a}_y + A\hat{a}_z)^2}$$

$$\frac{A_x^2 + A_y^2 + A_z^2}{A_x^2 + A_y^2 + A_z^2} = 1$$

6- $\vec{x} = 2\hat{a}_R + \frac{\pi}{4}\hat{a}_\theta + \frac{\pi}{2}\hat{a}_\phi$

$|\vec{x}| = \left(A\hat{a}_R^2 + A\hat{a}_\theta^2 + A\hat{a}_\phi^2\right)^{1/2} = \sqrt{2^2 + \left(\frac{\pi}{4}\right)^2 + \left(\frac{\pi}{2}\right)^2} =$

$\boxed{|\vec{x}| = \sqrt{4 + \frac{5\pi^2}{16}}}$

7- a) $\vec{A}\cdot\vec{B} = |\vec{A}|\cdot|\vec{B}|\cos(\theta)$

$\begin{cases} \vec{A} = 20\hat{a}_x + 15\hat{a}_y - 10\hat{a}_z \\ \vec{B} = \hat{a}_x + \hat{a}_y \end{cases}$

$|\vec{A}| = \sqrt{20^2 + 15^2 + (-10)^2} = 5\sqrt{29}$

$|\vec{B}| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\cos(\theta) = \frac{\vec{A}\cdot\vec{B}}{|\vec{A}|\cdot|\vec{B}|} = \frac{35}{\sqrt{2}\cdot 5\sqrt{29}} = \frac{7}{\sqrt{58}}$

$\vec{A}\cdot\vec{B} = 20\cdot 1 + 15\cdot 1 + (10)\cdot 0$

$\vec{A}\cdot\vec{B} = 35$

$\theta = \arccos\left(\frac{7}{\sqrt{58}}\right) =$

$\vec{A}\cdot\vec{B} = 5\sqrt{29}\cdot\sqrt{2}\cdot\frac{7}{\sqrt{58}} = \frac{35\sqrt{\cancel{58}}}{\sqrt{\cancel{58}}} = 35$

b) $\vec{A}\times\vec{B}$

$\vec{A}\times\vec{B} = \begin{vmatrix} a_x & a_y & a_z \\ 20 & 15 & -10 \\ 1 & 1 & 0 \end{vmatrix} \begin{matrix} a_x & a_y \\ 20 & 15 \\ 1 & 1 \end{matrix}$

$\boxed{\vec{A}\times\vec{B} = 10\hat{a}_x - 10a_y - 5\hat{a}_z}$

$(0)\hat{a}_x + (-10)a_y + 20\hat{a}_z - \left((0)\hat{a}_y + (-10)\hat{a}_x + 15\hat{a}_z\right)$

$10\hat{a}_x - 10a_y + 5\hat{a}_z$

c) $\text{Proy}_B A = \frac{\vec{A}\cdot\vec{B}}{|B|^2}\vec{B}$

$\vec{A}\cdot\vec{B} = 35$

$|B| = \sqrt{2}$, $|B|^2 = 2$

$= \frac{35}{2}(\hat{a}_x + \hat{a}_y) = \left\langle \frac{35}{2}\hat{a}_x + \frac{35}{2}\hat{a}_y \right\rangle$

8- $(A \cdot B)^2 = (A_x B_x + A_y B_y + A_z B_z)^2 = (A_x B_x + A_y B_y + A_z B_z)(A_x B_x + A_y B_y + A_z B_z)$

$= A_x^2 B_x^2 + 2 A_x A_y B_x B_y + 2 A_x A_z B_x B_z + A_y^2 B_y^2 + 2 A_y A_z B_y B_z + A_z^2 B_z^2$

$$|\vec{A} \times \vec{B}|^2 = \begin{vmatrix} a_x & b_y & a_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y)^2 + (A_z B_x - A_x B_z)^2 + (A_x B_y - A_y B_x)^2$$

$= A_y^2 B_z^2 - 2 A_y A_z B_z B_y + A_z^2 B_y^2 + A_z^2 B_x^2 - 2 A_z A_x B_x B_z + A_x^2 B_z^2 + A_x^2 B_y^2 -$
$2 A_x A_y B_y B_x + A_y^2 B_x^2$

$(AB)^2 = |A|^2 \cdot |B|^2 = (A_x^2 + A_y^2 + A_z^2)(B_x^2 + B_y^2 + B_z^2)$

$= A_x^2 B_x^2 + A_x^2 B_y^2 + A_x^2 B_z^2 + A_y^2 B_x^2 + A_y^2 B_y^2 + A_y^2 B_z^2 + A_z^2 B_x^2 + A_z^2 B_y^2 + A_z^2 B_z^2$

$(A \cdot B)^2 + |A \times B|^2 = (AB^2)$

$A_x^2 B_x^2 + 2 A_x A_y B_x B_y + 2 A_x A_z B_x B_z + A_y^2 B_y^2 + 2 A_y A_z B_y B_z + A_z^2 B_z^2$

$+ A_y^2 B_z^2 - 2 A_y A_z B_z B_y + A_z^2 B_y^2 + A_z^2 B_x^2 - 2 A_z A_x B_x B_z + A_x^2 B_z^2 + A_x^2 B_y^2$
$- 2 A_x A_y B_y B_x + A_y^2 B_x^2$

$= A_x^2 B_x^2 + A_x^2 B_y^2 + A_x^2 B_z + A_y^2 B_x^2 + A_y^2 B_y^2 + A_y^2 B_y^2 + A_z^2 B_z^2 + A_z^2 B_y^2 + A_z^2 B_z^2$

$= (AB)^2$

9- $\vec{P} = \cos\theta_1 \hat{a}_x + \operatorname{sen}\theta_1 \hat{a}_y$, $\vec{Q} = \cos\theta_2 \hat{a}_x + \operatorname{sen}\theta \hat{a}_y$

$|\vec{P}| = \sqrt{(\cos\theta_1)^2 + \operatorname{sen}^2\theta_1} = \sqrt{1} = 1$

$|\vec{Q}| = \sqrt{\cos^2\theta_2 + \operatorname{sen}^2\theta_2} = \sqrt{1} = 1$

Logo $\vec{P}$ e $\vec{Q}$ são vetores unitarios pois possuem magnitude igual a 1.

b) $\cos(\theta_2 + \theta_1) = \cos\theta_2 \cdot \cos\theta_1 - \operatorname{sen}\theta_2 \operatorname{sen}\theta_1$

$\cos(\theta_2 - \theta_1) = \cos\theta_2 \cdot \cos\theta_1 + \operatorname{sen}\theta_2 \operatorname{sen}\theta_1$

10- $\vec{OP} = P - O = (4, -5, 1) - (0,0,0) = \langle 4, -5, 1 \rangle$

$|\vec{OP}| = \sqrt{4^2 + (-5)^2 + 1^2} = \sqrt{42}$

$U_{OP} = \dfrac{\vec{OP}}{|\vec{OP}|} = \dfrac{\langle 4, -5, 1 \rangle}{\sqrt{42}} = \left\langle \dfrac{4}{\sqrt{42}}, \dfrac{-5}{\sqrt{42}}, \dfrac{1}{\sqrt{42}} \right\rangle$